OonR3: Selective Bispectrum for the Octahedral Group

Implementing the O-Bispectrum and solving the SO(3) inversion problem

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The octahedral group O

O is the group of 24 rotational symmetries of the cube.

Generators:

- $R_z(\pi/2)$: face rotation
- $R_{[1,1,1]}(2\pi/3)$: vertex rotation

5 real irreps (dims 1, 3, 3, 2, 1):

- ρ_0 (A_1): trivial
- ρ_1 (T_1): standard 3D rotation matrices
- ρ_2 (T_2): $\rho_4(g) \cdot \rho_1(g)$
- ρ_3 (E): 2D, involves $\pm\frac{1}{2}$, $\pm\frac{\sqrt{3}}{2}$
- ρ_4 (A_2): alternating (parity of body-diagonal permutation)

Selective bispectrum (App. B, Mataigne et al.):

4 matrix coefficients:

Pair	Size	Scalars
β_{ρ_0,ρ_0}	1×1	1
β_{ρ_0,ρ_1}	3×3	9
β_{ρ_1,ρ_1}	9×9	81
β_{ρ_1,ρ_2}	9×9	81
Total		**172**

Compare: full bispectrum $\mathcal{O}(|G|^2)$ = 576

Development overview

1. **Group data generation** — BFS from generators, verified against `escnn`
2. **Irrep matrices** — 3 constructed analytically, 1 from products, 1 hardcoded
3. **CG matrices** — projection operator method (no scipy)
4. **Forward pass** — non-commutative bispectrum formula
5. **DFT/inverse DFT** — left-convention, verified equivariance
6. **Inversion** — the hard part (3 slides)

Files:

- `src/bispectrum/o_on_r3.py` — 753 lines, the full module
- `tests/test_o_on_r3.py` — 68 tests
- `docs/o_on_r3_implementation_notes.md`

Group data: 24 elements as hardcoded tensors

Challenge: need exact integer 3×3 rotation matrices, Cayley table, inverse map, Kronecker table — all self-consistent, independent of `escnn` at runtime.

Approach:

1. BFS from generators $\langle R_z(\pi/2),\ R_{[1,1,1]}(2\pi/3)\rangle$
2. Round to integers, sort lexicographically $\Rightarrow$ canonical ordering
3. Cayley table: $C[i,j] =$ index of $g_i \cdot g_j$
4. Verified against `escnn.group.octa_group()` (element-wise match)

Identity element: g_{23} (not g_0) in lexicographic order.

All data embedded as literal `torch.tensor` constants — zero I/O at import.

CG matrices via projection operators

Need orthogonal C such that $C^\top(\rho_i(g) \otimes \rho_j(g))\, C = \bigoplus_\rho \rho(g)$ for all $g \in O$.

Method (projection onto irrep subspaces):

$$P_k = \frac{d_k}{|G|} \sum_{g \in G} \chi_k(g)^* \left[\rho_i(g) \otimes \rho_j(g)\right]$$

Eigendecompose P_k; eigenvectors with eigenvalue 1 span the ρ_k subspace.
Concatenate across all k in the decomposition $\Rightarrow C$.

Verification: for all 24 group elements, $C^\top(\rho_i \otimes \rho_j)C$ has zero off-diagonal blocks to machine precision ($< 10^{-10}$).

Two CG matrices needed:

- C_{11}: $\rho_1 \otimes \rho_1 = \rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3 \quad (9 \times 9)$
- C_{12}: $\rho_1 \otimes \rho_2 = \rho_1 \oplus \rho_2 \oplus \rho_3 \oplus \rho_4 \quad (9 \times 9)$

DFT convention: getting the signs right

Bug encountered: initial implementation used $\rho_k(g)^\top$ in the DFT, giving *right*-equivariance: $\mathcal{F}(T_h f)_k = \mathcal{F}(f)_k \cdot \rho_k(h)^\top$.

The bispectrum formula requires *left*-equivariance: $\mathcal{F}(T_h f)_k = \rho_k(h) \cdot \mathcal{F}(f)_k$.

Correct DFT:

$$\mathcal{F}(\rho_k) = \sum_{g \in O} f(g) \cdot \rho_k(g) \qquad \text{(no transpose)}$$

Correct inverse DFT:

$$f(g) = \frac{1}{|G|} \sum_k d_k \operatorname{tr}\big(\rho_k(g)^\top \mathcal{F}_k\big)$$

Effect of the bug: invariance error $\sim 10^2 \to \sim 10^{-13}$ after fix.

The Inversion Problem

Why the D_n bootstrap doesn't work for O

and what we did about it

Recap: bootstrap inversion for D_n (Algorithm 4)

For D_n, inversion is a clean sequential bootstrap:

1. $\mathcal{F}_0 = (\beta_{\rho_0,\rho_0})^{1/3}$
2. $\mathcal{F}_1^\top \mathcal{F}_1 = \beta_{\rho_0,\rho_1}/\mathcal{F}_0 \quad \Rightarrow \quad \mathcal{F}_1 = \sqrt{M}$ (symmetric square root, up to $O(2)$)
3. Extract remaining $\mathcal{F}_k$ from CG block-diagonalisation:

$$\bigoplus_\rho \mathcal{F}_\rho^\top = C^\top \beta_{\rho_1,\rho_{k-1}} \left[\mathcal{F}_1 \otimes \mathcal{F}_{k-1}\right]^{-1} C$$

Why it works: for 2D irreps, the $O(2)$ ambiguity Q in step 2 satisfies $C^\top(Q \otimes Q)C =$ block-diagonal for *all* $Q \in \mathrm{SO}(2)$, because the CG decomposition of 2D rotations respects the continuous group.

The ambiguity cancels perfectly during extraction. No leakage between irreps.

The SO(3) phase problem for O

For O, step 2 gives $\mathcal{F}_1 = S = \sqrt{\mathcal{F}_1^\top \mathcal{F}_1}$ (symmetric PSD, 3×3).

True $\mathcal{F}_1 = Q \cdot S$ for some $Q \in \mathrm{SO}(3)$. The extraction becomes:

$$C^\top \beta_{11}(S \otimes S)^{-1} C = \left(\bigoplus \mathcal{F}_k^\top\right) \cdot \underbrace{C^\top (Q \otimes Q) C}_{D}$$

The problem: D is block-diagonal in the SO(3) CG sense $(1+3+5)$, *not* the O CG sense $(1+3+3+2)$.

Under SO(3): $3 \otimes 3 = \ell{=}0 \oplus \ell{=}1 \oplus \ell{=}2 \quad (1+3+5)$.

Under O: $\rho_1 \otimes \rho_1 = \rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3 \quad (1+3+3+2)$.

The $\ell{=}2$ sector (dim 5) splits into ρ_2 (dim 3) $\oplus$ ρ_3 (dim 2) under O, but $D^{(2)}(Q)$ is a *full* 5×5 rotation for general Q. $\Rightarrow$ **Cross-talk** between ρ_2 and ρ_3 in the extraction.

What we verified numerically

Defined $R = C^\top \beta_{11}(S \otimes S)^{-1} C = (\bigoplus \mathcal{F}_k^\top) \cdot D$.

Quantity	Value	Interpretation
$R\,R^\top$ cross-block ($\rho_0 - \rho_1, \rho_2 - \rho_3$)	$\sim 10^{-13}$	$R\,R^\top$ *is* block-diagonal
$R\,R^\top[4{:}7,\ 4{:}7] - \mathcal{F}_2^\top \mathcal{F}_2$	$\sim 10^{-13}$	Fourier **norms** exactly recoverable
$R\,R^\top[4{:}7,\ 7{:}9]$ (cross ρ_2/ρ_3)	$\sim 10^{-14}$	No cross-block in $R\,R^\top$
$R[4{:}7,\ 7{:}9]$ (cross-block of R itself)	~ 4.0	**Severe leakage** in R
$D[1{:}4,\ 4{:}9]$ (cross $\ell{=}1, \ell{=}2$)	0	$\ell{=}1$ and $\ell{=}2$ *never* mix
Q from $R[1{:}4,\ 1{:}4]$	$= I$	Q **cancels** from ρ_1 block

Key insight: $R\,R^\top = \bigoplus \mathcal{F}_k^\top \mathcal{F}_k$ — Fourier *norms* are exactly recoverable. But the *phases* (orientations) are lost.

Why Q is invisible in β_{11}

The extraction result $R = (\bigoplus \mathcal{F}_k^\top) \cdot D$ has structure:

$$R = \begin{bmatrix} \mathcal{F}_0 & 0 & 0 \\ 0 & \underbrace{S\,Q^\top \cdot Q}_{=S} & 0 \\ 0 & 0 & [\mathcal{F}_2^\top, \mathcal{F}_3^\top] \cdot D^{(2)}(Q) \end{bmatrix}$$

- ℓ=0 block: gives $\mathcal{F}_0$ (known). No Q dependence.
- ℓ=1 block: $\mathcal{F}_1^\top Q = (QS)^\top Q = SQ^\top Q = S$. Q **cancels identically.**
- ℓ=2 block: 5×5 matrix mixing ρ_2 and ρ_3 through $D^{(2)}(Q)$. This is the *only* place Q appears, but it's entangled with the unknown $\mathcal{F}_2, \mathcal{F}_3$.

$\Rightarrow$ No closed-form extraction of Q from β_{11} alone. This is a **fundamental mathematical obstruction**, not an implementation bug.

Solution: Bootstrap + Levenberg-Marquardt correction

Phase 1 — Bootstrap initialization (closed-form, exact norms):

1. $\mathcal{F}_0 = (\beta_{\rho_0,\rho_0})^{1/3}$
2. $S = \sqrt{\beta_{\rho_0,\rho_1}/\mathcal{F}_0}$ (symmetric sqrt)
3. $\mathcal{F}_k^\top \mathcal{F}_k$ from $R\,R^\top$ diagonal blocks $\Rightarrow$ symmetric sqrts
4. Inverse DFT $\Rightarrow f_{\text{init}}$

Result: correct Fourier **norms** for all 5 irreps, wrong **phases**.

Phase 2 — LM corrections (single linear solve per step):

5. Jacobian $J = \frac{\partial \beta}{\partial f}$ via forward-mode AD (172×24 matrix)
6. Solve $(J^\top J + \mu I)\,\delta f = J^\top(\beta_{\text{target}} - \beta_{\text{pred}})$
7. Adaptive μ: halve on success, $5\times$ on failure
8. Repeat for $n_{\text{corrections}}$ steps

Inversion convergence

Single sample (seed 42):

Steps	$\|\beta_{\text{rec}} - \beta\|_\infty$	Signal err
0 (bootstrap)	3×10^2	—
1	8×10^1	0.88
3	5×10^0	0.08
5	2×10^{-4}	$< 10^{-5}$
10	8×10^{-5}	$< 10^{-5}$

Quadratic convergence once in the right basin.

Batch of 8 (6 seeds, 4 restarts $\times$ 10 LM steps):

Seed	$\|\beta\|_\infty$ err	Signal err
42	2×10^{-5}	$< 10^{-5}$
123	2×10^{-5}	$< 10^{-5}$
0	3×10^{-5}	$< 10^{-5}$
314	2×10^{-5}	$< 10^{-5}$
7	9×10^{-2}	0.017
99	1×10^{-1}	0.041

All seeds converge. Hard cases need more restarts.

Sanborn & Miolane (2025): what the paper provides

Appendix D gives a general inversion algorithm for non-commutative groups:

1. Recover $\mathcal{F}_{\rho_0} = (\beta_{\rho_0,\rho_0})^{1/3}$
2. Recover $\mathcal{F}_\rho^\dagger \mathcal{F}_\rho = \beta_{\rho,\rho_0}/\mathcal{F}_{\rho_0}^\dagger$, then take the symmetric square root via SVD:

$$\mathcal{F}_\rho = U\Sigma V^\dagger \;\Rightarrow\; \mathcal{F}_\rho^\dagger \mathcal{F}_\rho = V\Sigma^2 V^\dagger \;\Rightarrow\; \mathcal{F}'_\rho = V\Sigma V^\dagger$$

   The paper declares: "we fix $UV^\dagger = I$", analogous to fixing $\phi = 0$ in the commutative case.
3. Extract remaining $\mathcal{F}_k$ from:

$$\bigoplus_\rho \mathcal{F}_\rho = C^{-1}\left(\mathcal{F}'_1 \otimes \mathcal{F}'_1\right)^{-\dagger} \beta_{\rho_1,\rho_1}\, C^{-\dagger}$$

   "where everything on the right-hand side is known."

This works for D_n (D_4, D_{16}). The paper also confirms completeness for O and identifies the 4 selective pairs via Kronecker table BFS.

The hidden assumption that breaks for O

Step 2 gives $\mathcal{F}_1' = V\Sigma V^\dagger$, but the true coefficient is $\mathcal{F}_1 = Q \cdot \mathcal{F}_1'$ with $Q = UV^\dagger \in \mathrm{SO}(d_1)$.

Plugging $\mathcal{F}_1'$ into step 3 instead of $\mathcal{F}_1$:

$$C^\top \beta_{11} (\mathcal{F}_1' \otimes \mathcal{F}_1')^{-\dagger} C = \underbrace{\Big(\bigoplus_k \mathcal{F}_k^\top\Big)}_{\text{what we want}} \cdot \underbrace{C^\top (Q \otimes Q)\, C}_{D(Q)}$$

For D_n ($d_1 = 2$, $Q \in \mathrm{SO}(2)$):

- $D(Q)$ is block-diagonal $\forall Q \in \mathrm{SO}(2)$
- D_n-CG matrices block-diagonalise all of $\mathrm{SO}(2)$, not just D_n
- Reading off diagonal blocks gives exact $\mathcal{F}_k$

For O ($d_1 = 3$, $Q \in \mathrm{SO}(3)$):

- $\mathrm{SO}(3)$: $3 \otimes 3 = \ell{=}0 \oplus \ell{=}1 \oplus \ell{=}2$ (1+3+5)
- O: same $= \rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3$ (1+3+3+2)
- $\ell{=}2$ block (dim 5) **splits** into $\rho_2 \oplus \rho_3$
- $D^{(2)}(Q)$ is full 5×5 — mixes ρ_2, ρ_3

Numerical evidence and consequences

Verified numerically for random signals on O:

Quantity	Value	Implication
$R\,R^\top$ cross-block (ρ_{0-1} vs ρ_{2-3})	10^{-13}	Fourier *norms* exact
$R[4{:}7,\ 7{:}9]$ (cross ρ_2/ρ_3)	$\sim$4.0	Severe leakage
Q from ρ_1 block of R	$= I$	Q cancels from ρ_1
Bispectrum roundtrip (bootstrap only)	10^{+8}	Catastrophically wrong

Key insight: $R\,R^\top = \bigoplus \mathcal{F}_k^\top \mathcal{F}_k$ always. *Norms* recoverable; *phases* are not.

What the paper should state:

- The bootstrap algorithm is **valid only when all irreps have** $d \leq 2$
- For $d \geq 3$ irreps (O, icosahedral, $\mathrm{SO}(3)$, ...), "fixing $UV^\dagger = I$" introduces $\mathrm{SO}(d)$ error that propagates to all subsequent steps
- Completeness $\neq$ algorithmic computability: the inverse *exists* but this algorithm doesn't find it

Comparison: D_n vs O inversion

	D_n	O
Irrep dimensions	1D, 2D	1D, 2D, 3D
Ambiguity group	$O(2)$	$\mathrm{SO}(3)$
CG respects continuous group?	**Yes** ($\mathrm{SO}(2)$)	**No** ($\mathrm{SO}(3) \neq O$)
Sector mixing	None	$\rho_2 \leftrightarrow \rho_3$ in ℓ=2
Bootstrap alone	✓ exact	× wrong phases
Correction needed	None	LM (linear solve per step)
Inversion type	Closed-form	Bootstrap + refinement

Root cause: for any group G with irreps of dimension $d \geq 3$, the symmetric square root of $\mathcal{F}_1^\top \mathcal{F}_1$ introduces an $\mathrm{SO}(d)$ ambiguity. This is harmless when $d = 2$ (CG respects $\mathrm{SO}(2)$) but fatal when $d = 3$ (CG does *not* respect $\mathrm{SO}(3)$).

Usage

```python
from bispectrum import OonR3

bsp = OonR3()              # selective=True, 172 coefficients
f = torch.randn(4, 24)     # batch of 4 signals on O
beta = bsp(f)               # (4, 172), complex64

# Invariance under any of the 24 group actions
from bispectrum.o_on_r3 import _CAYLEY, _INVERSE
g = 7                       # any group element
g_inv = _INVERSE[g].item()
f_shifted = f[:, _CAYLEY[g_inv]]
torch.allclose(bsp(f_shifted).real, beta.real, atol=1e-4)  # True

# Inversion (up to group action)
f_rec = bsp.invert(beta)   # (4, 24)
# f_rec matches f up to one of the 24 group permutations
```

Test coverage: 68 new tests

Group data (9 tests)

- 24 elements, all orthogonal, all det $= 1$
- Cayley table closure and identity
- Inverse map consistency
- Kronecker table dimensions and symmetry

Irreps (3 tests)

- Dimensions $(1, 3, 3, 2, 1)$
- Identity element $\to I$
- Homomorphism: $\rho_k(g)\rho_k(h) = \rho_k(gh)$ for all g, h

CG matrices (4 tests)

- Orthogonality of C_{11}, C_{12}
- Block-diagonalisation for all 24 elements

Forward pass (33 tests)

- Output shape, dtype, determinism
- **Invariance under all 24 group actions** (parametrised)
- Output is real-valued (imaginary $< 10^{-10}$)
- Different signals produce different bispectra

DFT (4 tests)

- Roundtrip: DFT $\to$ inverse DFT $=$ identity
- Equivariance: $\mathcal{F}(T_g f) = \rho_k(g)\mathcal{F}(f)$

Inversion (7 tests)

- Bispectrum roundtrip, signal recovery
- Output shape, real-valued
- Bootstrap: exact $\mathcal{F}_0$, exact $\|\mathcal{F}_k\|$

What I learned

1. **DFT conventions matter** — left vs right equivariance is a one-transpose difference that silently breaks invariance by $O(10^2)$
2. **The SO(3) phase problem is fundamental** — not a bug, not fixable by a cleverer CG basis. Any group with $d \geq 3$ irreps will face this.
3. $R\,R^\top$ **trick** — even when the CG extraction R is non-block-diagonal, $R\,R^\top$ always gives exact Fourier norms. This provides a strong initialisation.
4. **Gauss-Newton without line search diverges** — the bispectrum is cubic, so the linearisation is only locally valid. Adaptive LM damping is essential.
5. **Multi-start is necessary** — the cubic bispectrum landscape has multiple basins of attraction. A single bootstrap initialisation lands in the wrong basin for $\sim$30% of random signals.

Limitations & next steps

Limitations:

- Inversion speed: $\sim$2s per sample (Jacobian computed per-sample via `jacfwd`)
- `selective=False` (full bispectrum) not yet implemented
- `invert()` not differentiable w.r.t. input β (uses `detach`)

Next steps:

- Analytical Jacobian $\partial\beta/\partial f$ (avoids AD, enables batching)
- Full octahedral group O_h ($|O_h| = 48$, adds improper rotations)
- GPU benchmarks
- Wire into equivariant network pipeline (pooling layer)
- Investigate whether the LM inversion can be made differentiable via implicit differentiation at the fixed point